

COMP90042 NLP

L3: N-gram Language Models



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Language Models

- In NLP, one important application is to determine the quality of text
 - Why a sentence is more **fluent** than another
- E.g. in speech recognition:
 - *recognise speech* > *wreck a nice beach*
- How to measure ‘goodness’? By using **probabilities** estimated by language models
- Language model can also be used for **generation**
 - Forms the basis of chatGPT!

Write a short fictional news story where Donald Trump became a vegan



Donald Trump Shocks the World by Adopting Vegan Lifestyle

New York City, October 27, 2023 – In a move no one saw coming, former President Donald Trump has announced he is now a committed vegan, sparking widespread reaction from around the globe. The announcement was made during an exclusive dinner at Trump Tower, where Trump unveiled his new book, *The Art of the Meal: Winning at Plants*, co-written with celebrity chef and plant-based advocate Tabitha Brown.

Speaking to a stunned crowd of reporters and supporters, Trump declared, "I've always been about winning. And let me tell you, folks, eating plants? It's a big win. Tremendous, actually. No one thought I could do it, but look at me now—healthier, sharper, and more energetic than ever. I've seen the light. Frankly, steak was overrated. Sad!"

The former president, who was well-known for his fondness for fast food and ketchup-drenched steaks, credited his transformation to a recent experience at a Mar-a-Lago charity event where he accidentally sampled a vegan burger. "They told me it was a regular burger," Trump recounted. "I took one bite and said, 'This is the best burger I've ever had.' When they told me it was made of plants, I said, 'Fake News!' But it wasn't. Incredible work by the plant people—just tremendous technology."

Trump's pivot to veganism has already sparked a cultural phenomenon, with his brand-new "Make America Green Again" (MAGA) hats selling out within hours. Trump Hotels have also announced a forthcoming "Plant-Based Suite Experience," featuring 100% vegan room service menus curated by Michelin-starred chefs.

While many of his political allies were initially skeptical, figures such as Senator Lindsey Graham have embraced the change, with Graham claiming, "If Donald Trump can give up fried chicken, I can say goodbye to pork chops."

Not everyone is thrilled about the shift, however. Critics from Trump's loyalist base have taken to social media, with some accusing him of "selling out" to the plant-based movement. Far-right commentator Tucker Carlson called it "an unforgivable betrayal," while a famous BBQ chain in Texas announced it would refuse to serve Trump-branded plant-based products.

Animal rights groups, meanwhile, have celebrated the news as a surprising but welcome victory. PETA issued a statement praising Trump's transformation, calling it "a monumental step towards saving animals and the planet." Environmentalists have also lauded the move, noting that even a reduction in Trump's cheeseburger intake could significantly impact methane emissions nationwide.

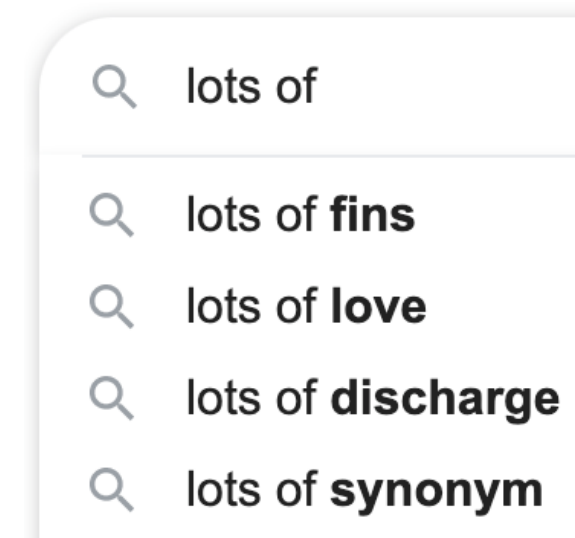
In typical Trump fashion, the former president has already hinted at a new business endeavor: a chain of luxury vegan restaurants called "Trump Gardens." According to his announcement, the restaurants will feature gold-plated salad tongs, menus printed on 100% recycled paper, and a signature dessert called the "Donald Double-Decker Dairy-Free Cheesecake."

"I've always been ahead of the curve," Trump said, glowing under the spotlight. "Some people lead by example. I'm leading by... eating beautifully. Believe me, folks, you're going to love plants. I do deals with plants like no one else."

As the world adjusts to a seemingly greener Trump, experts are left to wonder: Is this a fleeting rebranding effort, or the beginning of a plant-based legacy? Only time—and the contents of his next plate—will tell.

Language Models

- Useful for
 - Query completion
 - Optical character recog.
- Other generation tasks
 - Machine translation
 - Summarisation
 - Dialogue systems



Outline

- Deriving n -gram language models
- Smoothing to deal with sparsity

Probabilities: Joint to Conditional

Our goal is to get a probability for an arbitrary sequence of m words

$$P(w_1, w_2, \dots, w_m)$$

First step is to apply the chain rule to convert joint probabilities to conditional ones

$$P(w_1, w_2, \dots, w_m) = P(w_1)P(w_2 | w_1)P(w_3 | w_1, w_2) \dots \\ P(w_m | w_1, \dots, w_{m-1})$$

The Markov Assumption

Still intractable, so make a simplifying assumption:

$$P(w_i | w_1 \dots w_{i-1}) \approx P(w_i | w_{i-n+1} \dots w_{i-1})$$

For some small n

When $n = 1$, a unigram model

$$P(w_1, w_2, \dots, w_m) = \prod_{i=1}^m P(w_i)$$

the dog ^{w_i} barks

When $n = 2$, a bigram model

$$P(w_1, w_2, \dots, w_m) = \prod_{i=1}^m P(w_i | w_{i-1})$$

the ^{w_i} dog barks

When $n = 3$, a trigram model

$$P(w_1, w_2, \dots, w_m) = \prod_{i=1}^m P(w_i | w_{i-2} w_{i-1})$$

the ^{w_i} dog barks

Maximum Likelihood Estimation

How do we calculate the probabilities? Estimate based on counts in our corpus:

For unigram models,

$$P(w_i) = \frac{C(w_i)}{M} \quad \frac{C(\text{barks})}{M}$$

← Total number of word tokens in corpus

For bigram models,

$$P(w_i | w_{i-1}) = \frac{C(w_{i-1}w_i)}{C(w_{i-1})} \quad \frac{C(\text{dog barks})}{C(\text{dog})}$$

For n-gram models generally,

$$P(w_i | w_{i-n+1} \dots w_{i-1}) = \frac{C(w_{i-n+1} \dots w_i)}{C(w_{i-n+1} \dots w_{i-1})}$$

Book-ending Sequences

- Special tags used to denote start and end of sequence
 - `<s>` = sentence start
 - `</s>` = sentence end

Trigram example

Corpus:

yes no no no no yes
no no no yes yes yes no

What is the probability of

yes no no yes

under a trigram language model?

$P(\text{yes no no yes}) =$

$P(\text{yes} \mid \langle s \rangle \langle s \rangle) \times$

$P(\text{no} \mid \langle s \rangle \text{yes}) \times$

$P(\text{no} \mid \text{yes no}) \times$

$P(\text{yes} \mid \text{no no}) \times$

$P(\langle /s \rangle \mid \text{no yes})$

 Need to predict $\langle /s \rangle$ because it's the end of sentence!

Corpus:

<s> <s> yes no no no no yes </s>

<s> <s> no no no yes yes yes no </s>

Compute: $P(\textit{yes no no yes})$

$$P(w_i | w_{i-2}, w_{i-1}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

Corpus:

<s> <s> *yes* no no no no yes </s>

<s> <s> *no* no no yes yes yes no </s>

Compute: $P(\text{yes no no yes})$

$$P(w_i | w_{i-2}, w_{i-1}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

$P(\text{yes} | \text{<s> <s>})$

1/2

<s> <s> *yes*

<s> <s> *no*

Corpus:

<s> **<s> yes no** no no no yes </s>

<s> <s> no no no yes yes yes no </s>

Compute: $P(\text{yes no no yes})$

$$P(w_i | w_{i-2}, w_{i-1}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

$P(\text{yes} \text{<s> <s>})$	1/2	<s> <s> yes <s> <s> no
$P(\text{no} \text{<s> yes})$	1/1	<s> yes no

Corpus:

<s> <s> *yes no no* no no yes </s>
 <s> <s> no no no yes yes *yes no* </s>

Compute: $P(\text{yes no no yes})$

$$P(w_i | w_{i-2}, w_{i-1}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

$P(\text{yes} \text{<s> <s>})$	1/2	<s> <s> <i>yes</i> <s> <s> <i>no</i>
$P(\text{no} \text{<s> yes})$	1/1	<s> <i>yes no</i>
$P(\text{no} \text{yes no})$	1/2	<i>yes no no</i> <i>yes no </s></i>

Corpus:

<s> <s> yes no no no no yes </s>

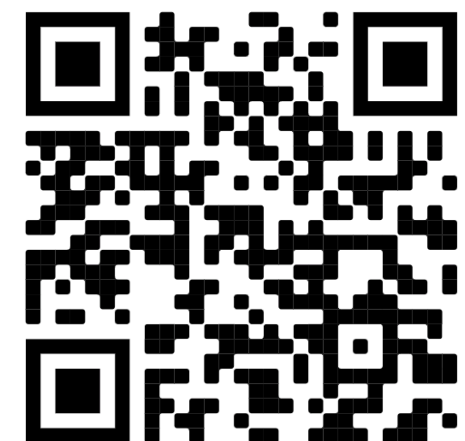
<s> <s> no no no yes yes yes no </s>

Compute: $P(\text{yes no no yes})$

$$P(w_i | w_{i-2}, w_{i-1}) = \frac{C(w_{i-2}, w_{i-1}, w_i)}{C(w_{i-2}, w_{i-1})}$$

$P(\text{yes} \text{<s> <s>})$	1/2	<i><s> <s> yes <s> <s> no</i>
$P(\text{no} \text{<s> yes})$	1/1	<i><s> yes no</i>
$P(\text{no} \text{yes no})$	1/2	<i>yes no no yes no </s></i>
$P(\text{yes} \text{no no})$	?	

[PollEv.com/jeyhanlau569](https://poll-ev.com/jeyhanlau569)



Several Problems

- Language has long distance effects — need large n
 - The *lecture/s* that took place last week *was/were* on preprocessing.
- Resulting probabilities are often very small
 - Use log probability to avoid numerical underflow
- What about unseen n -grams?
 - $P(w_1, w_2, \dots, w_m) = P(w_1 | \langle s \rangle) \times P(w_2 | w_1) \dots$
 - whole term = 0 if $P(w_2 | w_1) = 0$
↓
 - Need to smooth the LM!

Smoothing

Smoothing

- Basic idea: give events you've never seen before some probability
- Must be the case that $P(\text{everything}) = 1$
- Many different kinds of smoothing
 - Laplacian (add-one) smoothing
 - Add- k smoothing
 - Absolute discounting
 - Kneser-Ney
 - And others...

Laplacian (Add-one) Smoothing

- Simple idea: pretend we've seen each n -gram once more than we did.

For unigram models (V = the vocabulary),

$$P_{add1}(w_i) = \frac{C(w_i) + 1}{M + |V|}$$

For bigram models,

$$P_{add1}(w_i | w_{i-1}) = \frac{C(w_{i-1}w_i) + 1}{C(w_{i-1}) + |V|}$$

Add-one Example

<s> the rat ate the cheese </s>

What's the bigram probability $P(\text{ate} \mid \text{rat})$ under add-one smoothing?

$$= \frac{C(\text{rat ate}) + 1}{C(\text{rat}) + |V|} = \frac{2}{6} \quad V = \{ \text{the, rat, ate, cheese, } \langle /s \rangle \}$$

What's the bigram probability $P(\text{ate} \mid \text{cheese})$ under add-one smoothing?

$$= \frac{C(\text{cheese ate}) + 1}{C(\text{cheese}) + |V|} = \frac{1}{6}$$

<s> is not part of vocabulary because we never need to infer its conditional probability (e.g. $P(\langle s \rangle \mid \dots)$)

Recall: $P(\text{yes no no yes}) =$
 $P(\text{yes} \mid \langle s \rangle \langle s \rangle) \times P(\text{no} \mid \langle s \rangle \text{yes}) \times$
 $P(\text{no} \mid \text{yes no}) \times P(\text{yes} \mid \text{no no}) \times$
 $P(\langle /s \rangle \mid \text{no yes})$

Add- k Smoothing

- Adding one is often too much
- Instead, add a fraction k
- AKA Lidstone Smoothing, or Add- α Smoothing

$$P_{addk}(w_i | w_{i-1}, w_{i-2}) = \frac{C(w_{i-2}, w_{i-1}, w_i) + k}{C(w_{i-2}, w_{i-1}) + k |V|}$$

- Have to choose k

Lidstone Smoothing

Context = *alleged*

- 5 observed bi-grams
- 2 unobserved bi-grams

			Lidstone smoothing, $\alpha = 0.1$	
	counts	unsmoothed probability	effective counts	smoothed probability
<i>impropriety</i>	8	0.4	7.826	0.391
<i>offense</i>	5	0.25	4.928	0.246
<i>damage</i>	4	0.2	3.961	0.198
<i>deficiencies</i>	2	0.1	2.029	0.101
<i>outbreak</i>	1	0.05	1.063	0.053
<i>infirmity</i>	0	0	0.097	0.005
<i>alleged</i>	0	0	0.097	0.005
	20	1.0	20	1.0

$$0.391 \times 20$$

$$(0 + 0.1) / (20 + 7 \times 0.1)$$

$$(8 + 0.1) / (20 + 7 \times 0.1)$$

Absolute Discounting

- ‘Borrows’ a **fixed** probability mass from observed n-gram counts
- Redistributes it to unseen n-grams

Absolute Discounting

Context = *alleged*

- 5 observed bi-grams
- 2 unobserved bi-grams

			Lidstone smoothing, $\alpha = 0.1$		Discounting, $d = 0.1$	
	counts	unsmoothed probability	effective counts	smoothed probability	effective counts	smoothed probability
<i>impropriety</i>	8	0.4	7.826	0.391	7.9	0.395
<i>offense</i>	5	0.25	4.928	0.246	4.9	0.245
<i>damage</i>	4	0.2	3.961	0.198	3.9	0.195
<i>deficiencies</i>	2	0.1	2.029	0.101	1.9	0.095
<i>outbreak</i>	1	0.05	1.063	0.053	0.9	0.045
<i>infirmity</i>	0	0	0.097	0.005	0.25	0.013
<i>alleged</i>	0	0	0.097	0.005	0.25	0.013
	20	1.0	20	1.0	20	1.0

$8 - 0.1$
 $(0.1 \times 5) / 2$
 $0.25 / 20$

Backoff

- Absolute discounting redistributes the probability mass **equally** for all unseen n-grams
- Katz Backoff: redistributes the mass based on a **lower order model** (e.g. unigram)

$$P_{katz}(w_i|w_{i-1}) = \begin{cases} \frac{C(w_{i-1}, w_i) - D}{C(w_{i-1})}, & \text{if } C(w_{i-1}, w_i) > 0 \\ \alpha(w_{i-1}) \times \frac{P(w_i)}{\sum_{w_j: C(w_{i-1}, w_j)=0} P(w_j)}, & \text{otherwise} \end{cases}$$

sum unigram probabilities for all words that do not co-occur with context w_{i-1}
 e.g. $P(\text{infirmity}) + P(\text{alleged})$

unigram probability for w_i
 e.g. $P(\text{infirmity})$

the amount of probability mass that has been discounted for context w_{i-1}
 $((0.1 \times 5) / 20 \text{ in previous slide})$

Issues with Katz Backoff

$$P_{katz}(w_i|w_{i-1}) = \begin{cases} \frac{C(w_{i-1}, w_i) - D}{C(w_{i-1})}, & \text{if } C(w_{i-1}, w_i) > 0 \\ \alpha(w_{i-1}) \times \frac{P(w_i)}{\sum_{w_j: C(w_{i-1}, w_j)=0} P(w_j)}, & \text{otherwise} \end{cases}$$

- *I can't see without my reading ____*
- $C(\text{reading}, \text{glasses}) = C(\text{reading}, \text{Francisco}) = 0$
- $C(\text{Francisco}) > C(\text{glasses})$
- Katz backoff will give higher probability to *Francisco*

Kneser-Ney Smoothing

- Redistribute probability mass based on the **versatility** of the lower order n-gram
- AKA “continuation probability”
- What is versatility?
 - High versatility -> co-occurs with a lot of unique words, e.g. glasses
 - men’s glasses, black glasses, buy glasses, etc
 - Low versatility -> co-occurs with few unique words, e.g. francisco
 - san francisco

Kneser-Ney Smoothing

$$P_{KN}(w_i | w_{i-1}) = \begin{cases} \frac{C(w_{i-1}, w_i) - D}{C(w_{i-1})}, & \text{if } C(w_{i-1}, w_i) > 0 \\ \beta(w_{i-1})P_{cont}(w_i), & \text{otherwise} \end{cases}$$

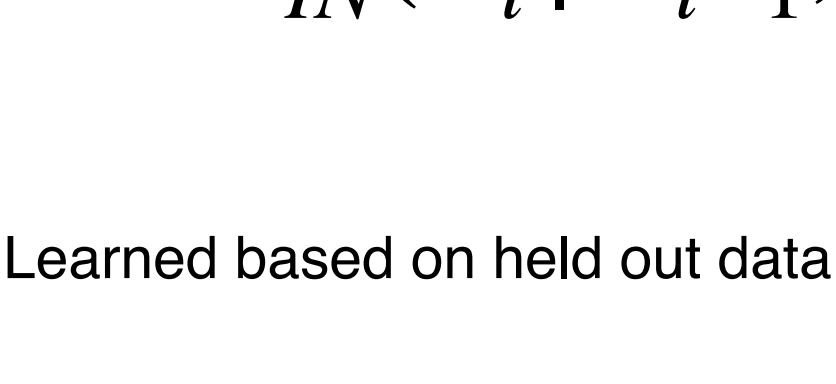
the amount of probability mass that has been discounted for context w_{i-1}

$$P_{cont}(w_i) = \frac{|\{w'_{i-1} : C(w'_{i-1}, w_i) > 0\}|}{\sum_{\{w_j : C(w_{i-1}, w_j) = 0\}} |\{w_{j-1} : C(w_{j-1}, w_j) > 0\}|}$$

- Intuitively the numerator counts #unique preceding words of w_i
- Denominator sums all continuation counts for unseen bigrams
- High continuation counts for glasses
- Low continuation counts for Franciso

Interpolation

- A better way to combine different orders of n -gram models
- Interpolated trigram model:

$$P_{IN}(w_i | w_{i-1}, w_{i-2}) = \lambda_3 P_3(w_i | w_{i-2}, w_{i-1}) \\ + \lambda_2 P_2(w_i | w_{i-1}) \\ + \lambda_1 P_1(w_i)$$


Learned based on held out data

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

A Final Word

- *N*-gram language models are a simple but effective way to capture the predictability of language
- Can be trained in an unsupervised fashion, scalable to large corpora
- Require smoothing to be effective
- Modern language models uses neural networks (lecture 7)

Readings

- E18 Chapter 6 (skip 6.3)